Birzeit University

Experiment #2

Source Internal Resistance,

Loading Problems And Circuit Impedance Matching

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Aim of the exp:

The aim of this experiment was to determine the load resistance (RL), while taking into account the internal resistance (Rin) of the source. Specifically, we aimed to find the value of RL that would satisfy the condition RL = R + Rin, where R is the resistance of the load resistor.

Results and calculation:

The experiment showed that the value of the load resistor that maximizes power transfer is influenced by the internal resistance of the voltage source. The practical value of the load resistor was found to be different from the theoretical value due to the presence of the internal resistance. Additionally, the graphs of power versus resistance and reciprocal of current versus resistance showed a clear trend, with the former being a semi-log I don know asemi log graph and the latter being a linear graph

The slope from the R and 1/I equal to 0.1039

E= 1/slope

E= 9.6 Volts And thy y-intercept is 0.111

Then R=y*E == 0.111*9.6 =1.066 Kohm

Then the r(in) =66 ohm

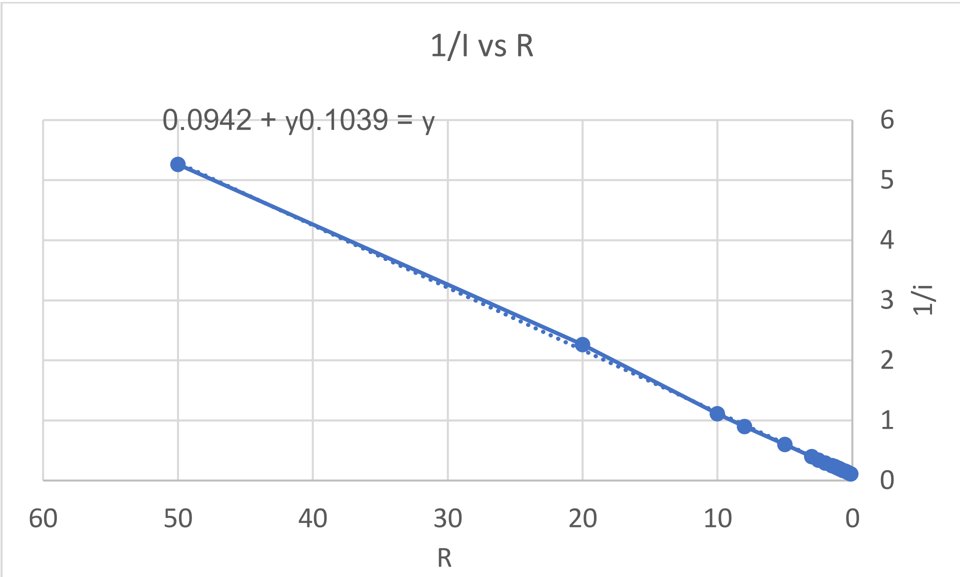
The max power was when the Resistance is 1.015 Kohms

And it was 24.9 watt.

From the graph. RL=Rin+R=1010 ohm R=1000 ohm (from graph) so the difference is 15 ohms

R(L)	I(mA)	1/I	Ipow(2)	Ipow(2)*R(l)=P(l)
0.1	9	0.11111	81	81
0.3	7.63	0.131	58.21	17.46
0.5	6.62	0.151	43.82	21.91
0.7	5.85	0.17	34.22	23.954
0.9	5.24	0.19	27.45	24.705
1	4.99	0.2	24.9	24.9
1.1	4.75	0.21	22.56	24.816
1.3	4.34	0.23	18.83	24.479
1.5	4	0.25	16	24
2	3.34	0.29	11.15	22.3
2.5	2.86	0.34	8.17	20.425
3	2.5	0.4	6.25	18.75
5	1.67	0.6	2.78	13.9
8	1.11	0.9	1.2321	9.85
10	0.9	1.111	0.81	8.1
20	0.27	2.263	0.22	4.4
50	0.19	5.263	0.036	1.805

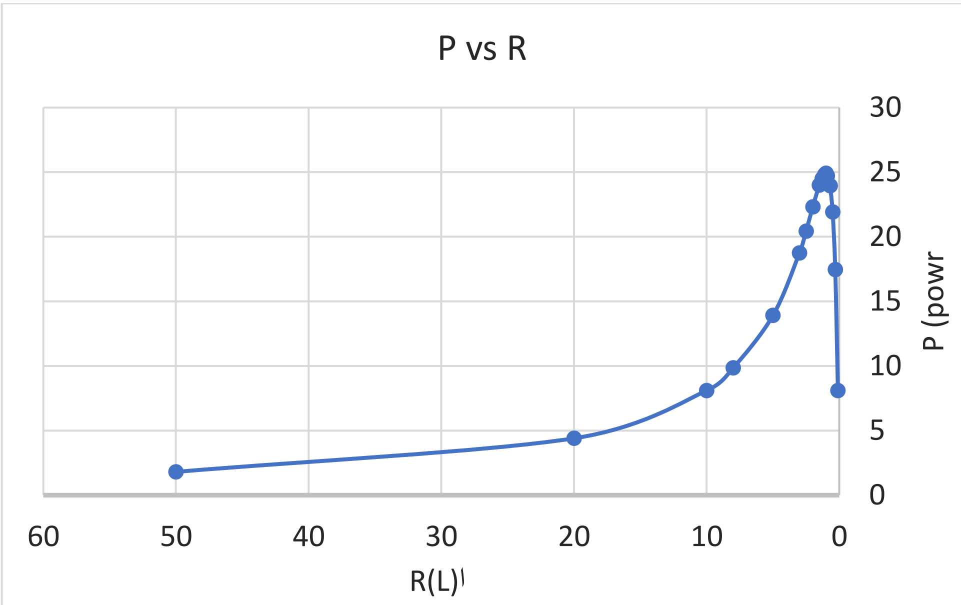
This diagram shows the relation between 1/I and the resistance:



The slope is 0.1039

And the thy y-intercept is 0.111

This diagram indicates the relation between the power in mwatt and the resistors:



The max of power is 24.9

When r=1.015

Results & Conclusion: • There should be an internal resistance in every circuit. • To reach the maximum value of the power transferring the load resistance should be equal to the sum of the additional resistance and the internal resistance. • There were some percentages of error in finding the load resistance that satisfies the condition of maximum power transfer due to systemic errors and random errors